

Customer Shopping Behavior Analysis

Uncovering insights from **3,900 transactions** across product categories to guide strategic decisions.



Dataset at a Glance

3,900

Purchases

18

Features

37

Missing Values

Review Rating column

\$59.76

Avg. Purchase

Demographics, purchase details, and shopping behavior features — loaded into PostgreSQL for analysis.

Data Preparation in Python

Cleaning

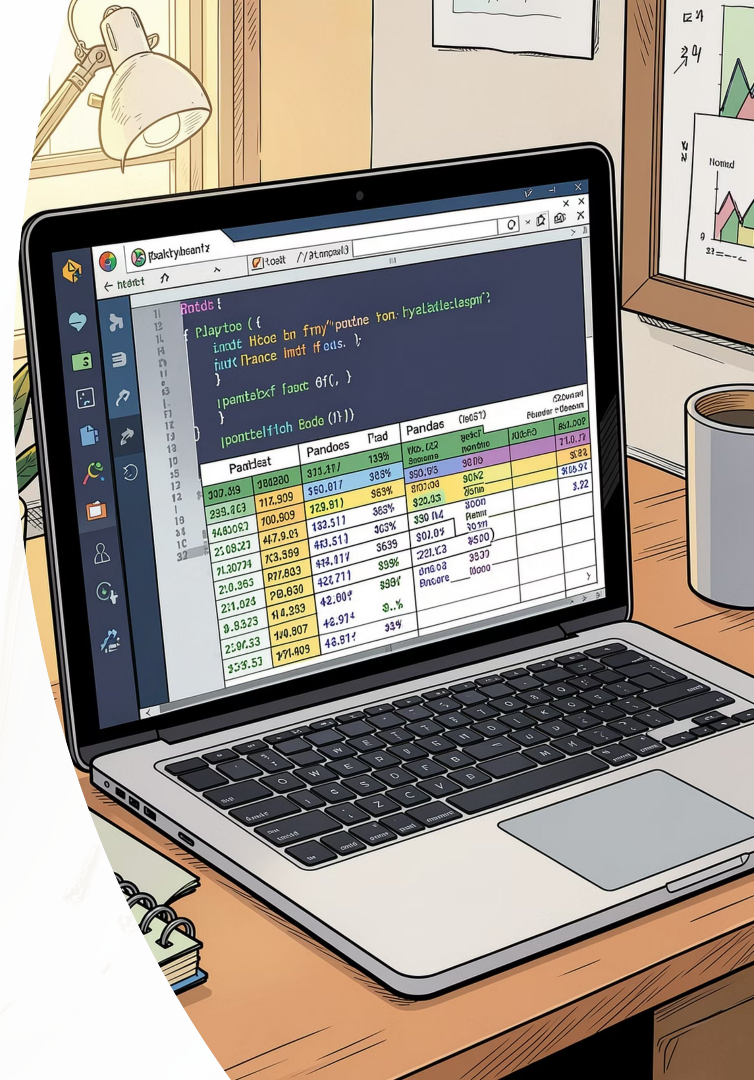
Imputed missing Review Ratings using median by category

Feature Engineering

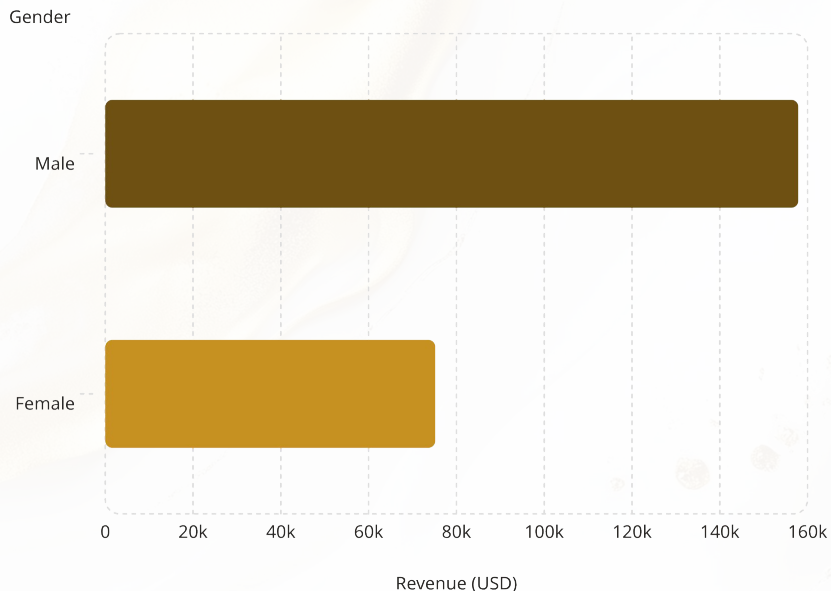
Created **age_group** and **purchase_frequency_days** columns

Integration

Loaded cleaned DataFrame into **PostgreSQL**



Revenue by Gender



Male customers drive **2× more revenue**

Male: **\$157,890**

Female: **\$75,191**

Male customers also dominate the non-subscriber segment — 2,518 vs. 958 repeat buyers without subscriptions.

Top Products & Discount Patterns

Top 5 Products by Rating

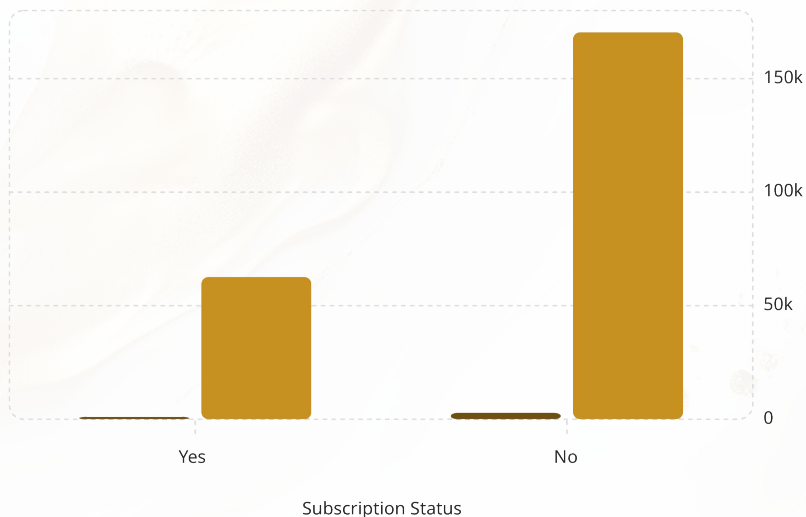
Item	Avg Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

Most Discounted Products

Item	Discount Rate
Hat	50%
Sneakers	49.66%
Coat	49.07%
Sweater	48.17%
Pants	47.37%

Subscribers vs. Non-Subscribers

■ Total Customers ■ Total Revenue (USD)



Non-subscribers = 73% of customers

Yet average spend is nearly identical:

Subscribers: **\$59.49**

Non-subscribers: **\$59.87**

- 958 repeat buyers are subscribers — opportunity to convert the 2,518 non-subscribing repeat buyers.



Customer Segmentation

🏆 Loyal

3,116 customers

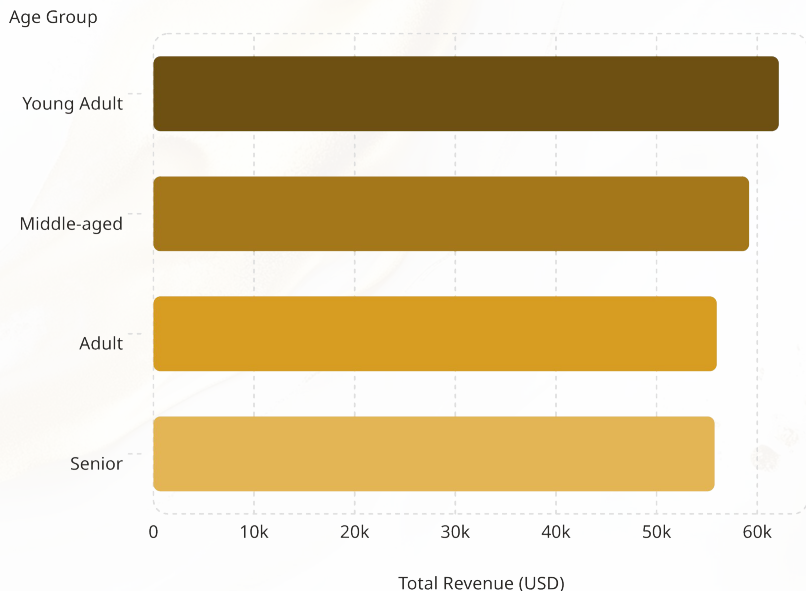
↺↻ Returning

701 customers

NEW New

83 customers

Revenue by Age Group



Young Adults lead revenue

All four age groups contribute relatively evenly — **no single segment dominates**.

Express shipping users spend slightly more on average (**\$60.48**) vs. Standard (**\$58.46**).

Power BI Dashboard



3.9K Customers



\$59.76 Avg Purchase



3.75 Avg Rating



Interactive Filters

Subscription, Gender, Category, Shipping



Business Recommendations

- **Boost Subscriptions**
Promote exclusive subscriber benefits
- **Loyalty Programs**
Reward repeat buyers toward "Loyal" segment
- **Review Discount Policy**
Balance sales lift with margin control
- **Targeted Marketing**
Focus on high-revenue age groups & express-shipping users

